

Maitreya Mohapatra

AI Engineer

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📅 Available: full-time from Oct 2026

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🇮🇳 Nationality: Indian

🐙 GitHub: github.com/mimo1999

🌐 Portfolio: mimo1999.github.io/



Professional Summary

AI Engineer (3+ yrs) building production agentic/RAG systems and CI/CD-deployed ML services across fintech, health-care, and industrial R&D. Co-inventor, 2 US patents. Completing M.Sc. AI at FAU Erlangen-Nürnberg.

Work Experience

AI Infra Engineer (Working Student) | Siemens Healthineers | Erlangen, Germany

04/2025 – present

Python, Azure DevOps, Slurm, Podman, Linux, Pytest

- Designed and implemented codebase-wide RAG ingestion across 30+ FPGA repositories, converting repository dependencies and code structure into knowledge graphs. Exposed the retrieval layer through MCP to a Copilot agent, enabling skill-based RTL orchestration and Questa OneSpin validation workflows.
- Cut CI runtime ~20% and manual effort ~60% by redesigning Azure DevOps CI/CD into a parallel execution framework.
- Improved compute utilization ~50% and cut turnaround ~35% by architecting a hybrid Azure DevOps-Slurm pipeline for FuseSoC.

Associate Software Engineer, Data Science | HighRadius | Bhubaneswar, India

06/2021 – 08/2023

Python, SQL, Snowflake, Jenkins, Docker, AWS

- Led ML treasury-forecasting delivery for 8 Fortune 500 enterprise accounts as primary technical contact; mentored 10+ interns.
- Built two patented models: payroll cash-flow forecasting (95%+ accuracy) and a customer-behavior feature framework that lifted accuracy 20–30% across production accounts.
- Productized 50+ forecasting models processing up to 6 million line items into HighRadius's AI-powered Treasury product, recognized as an IDC MarketScape Major Player.
- Cut AWS costs ~20% via compute-efficient pipeline design, and client implementation time 10–15% by standardizing multi-ERP (SAP, Oracle, NetSuite) ingestion into SQL/Snowflake ETL pipelines.

Internships and Part-time Roles

- Teaching Assistant, Deep Learning | FAU Erlangen-Nürnberg | Erlangen, Germany** (11/2024 – 03/2025): Mentored M.Sc. students in implementing and debugging CNN, RNN, and Transformer models using PyTorch.
- Data Science Intern | HighRadius | Bhubaneswar, India** (06/2020 – 06/2021): Cut client-side forecast validation time 50% with an Openpyxl report, and manual forecast-loading effort 70% via cloud scripts.
- Full Stack Intern | HighRadius | Bhubaneswar, India** (04/2020 – 06/2020): Built an AI-enabled accounts-receivable dashboard (React, Java/SQL, Dialogflow) with an XGBoost payment-date prediction model.

Education

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)

10/2023 – 09/2026

Master of Science in Artificial Intelligence (*graduating September 2026*) • Grade: 1.6 (German scale)

Kalinga Institute of Industrial Technology, Bhubaneswar (KIIT)

06/2017 – 06/2021

Bachelor of Technology in Computer Science and Engineering • Grade: 1.4 (German scale equivalent)

Technical Skills

- AI Systems & LLM Engineering:** Agentic AI, Multi-Agent Systems, MCP, RAG, GraphRAG, Tool Calling, Structured Outputs, Context Engineering, Fine-Tuning (LoRA), OpenAI/Anthropic APIs, LangGraph, LangChain, HuggingFace, PyTorch, scikit-learn, OpenCV, Explainable AI (SHAP)
- LLM Evaluation & Observability:** Ragas, LangSmith, Langfuse, MLflow, LLM-as-Judge, Evaluation Pipelines, Benchmarking, Guardrails, Hallucination Control, Harness Engineering, LLM Inference, ONNX
- AI Infrastructure & Serving:** Distributed Systems, Docker, Kubernetes, Terraform, Linux, CI/CD (Azure DevOps, Jenkins, GitHub Actions), Slurm/HPC, FastAPI, CUDA, AWS/Azure/GCP, Pydantic
- Data & Retrieval:** PostgreSQL/pgvector, ChromaDB, Neo4j, MongoDB, Snowflake, Databricks, dbt, Hybrid Retrieval
- Programming:** Python, R, Go, SQL, C++, TypeScript/Javascript, Node, React, Java, Bash, Git, PyTest, PySpark

Languages

English (C2) • German (A2, actively improving) • Hindi (C2) • Bengali (C1)

Research, Patents & Projects

- Master's Thesis:** built a multimodal model fusing audio and high-speed videoendoscopy to detect laryngeal dystonia biomarkers, reaching 86% patient-level diagnostic accuracy; integrated into a clinical application.
- US Patents (HighRadius):** Cash-flow forecasting (granted, US-12657596-B2); transaction-pattern prediction (applied).
- Projects:** **Clinical RAG:** offline clinical-guideline RAG, hybrid retrieval + reranking (88.9% Recall@5, 100% refusal correctness); **Research Swarm:** LangGraph pipeline where a 2B local model does the research and a larger one only plans and writes, with quotes, citations and scope enforced in code; **Flowify:** GraphRAG code explorer that maps a change's blast radius (risk level, callers, affected modules) and serves it to IDE assistants over MCP.